Applying the 10 Fundamentals at Each Layer

Frontend (React Dashboard):
- Requirements – show threat data, alerts, hero panel.
- Scale – thousands of concurrent users.
- Architecture – React + WebSockets + chart libraries.
- Data – cache last 5 min in memory, IndexedDB for persistence.
- Processing – UI state management, real-time updates.
- Reliability – graceful fallback if WebSocket drops.
- Performance – <2s render, throttled updates.
- Security – JWT auth, tenant isolation in UI.
- Observability – client logs (Sentry), perf monitoring.
- Cost – lightweight libraries, CDN-delivered assets.

API Layer (REST / GraphQL / WebSockets):
- Requirements – provide live + historical data, control actions.
- Scale – millions of requests/sec.
- Architecture – API Gateway (Envoy/NGINX), GraphQL subscriptions.
- Data – pass-through or aggregate before sending.
- Processing – input validation, RBAC, throttling.
- Reliability – retries, rate limiting, canary deployments.
- Performance – <100ms response time.
- Security – OAuth/JWT, TLS, audit logs.
- Observability – API metrics (latency, error rate).
- Cost – auto-scale nodes, caching.

Backend (Event Processing + Services):
- Requirements – ingest, normalize, detect anomalies.
- Scale – billions of events/day.
- Architecture – Kafka + Flink + microservices.
- Data – streams + intermediate state (Redis).
- Processing – windowed aggregations, enrichment.
- Reliability – checkpoints, failover clusters.
- Performance – <1s anomaly detection.
- Security – service-to-service mTLS, RBAC.
- Observability – metrics, logs, traces.
- Cost – partitioning, autoscaling Flink jobs.

Database / Storage:
- Requirements – store hot, warm, cold data.
- Scale – TBs/day ingestion, PBs over years.
- Architecture – Redis (hot), ClickHouse (warm), S3 (cold).
- Data – partitioned by tenant, region, time.
- Processing – queries, indexing, compaction.
- Reliability – replication, snapshots, backups.
- Performance – <2s analytics queries, <10ms Redis reads.
- Security – encryption at rest, tenant isolation.
- Observability – slow query logs, DB metrics.
- Cost – tiered storage, lifecycle policies.